

One relation behind the calculators

$$\sigma_{ab}(i) \approx \sum_s I_s D_{ab}(\mathbf{r}_{is}), \quad D_{ab} = \partial_a \partial_b \frac{1}{\rho} = \frac{3\hat{\rho}_a \hat{\rho}_b - \delta_{ab}}{\rho^3}$$

each contributor is a named, equation-bearing projection of it — the question is whether ours correspond.

where $\sigma_{ab}(i)$ — the shielding tensor at nucleus i (what DFT gives, what we predict); I_s — how strongly source s contributes (its calibration coefficient, geometry to ppm); D_{ab} — the dipolar kernel for the nucleus-to-source step $\mathbf{r}_{is}$; ρ , $\hat{\rho}$ — that step's length and direction; $(3\cos^2\theta - 1)/\rho^3$ — strongest along the axis, zero at the magic angle ($\cos^2\theta = \frac{1}{3}$).

classes: ■ question ■ data source ■ evaluative tool ■ model ■ finding

A · distribution in space

Stage 1

- Are there probable shielding patterns across the backbone dihedrals? (a candidate category)

uses: 720, extractor + dihedral analysis $\rightarrow \phi/\psi$ shielding patterns

B · do the contributors correspond to a law?

— recognisable by name and equation Stage 1 $\rightarrow$ 2 · 720 = cross-protein

- How does the ridge fit among them?
- Do they reflect the physics — match their named equations?
- Can they operate as a model? (the D_{ab} -sum)
- Do they help predict the shift?

uses: 720, 1P9J, ridge, contribution, stats, D_{ab} -sum $\rightarrow$ which correspond

C · what has the model learned?

Stage 3

- Does the shielding model + the contributors help predict shift? (ablation)
- Does it favour secondary structure?
- Does it favour specific residues / atoms (IUPAC)?
- Does it favour specific regions within proteins?
- What geometric features bear on its response?

uses: BMRB, ablation, Model 1 / Model 2 $\rightarrow$ what it favours

the named projections

ring current — McConnell · Johnson–Bovey · Haigh–Mallion

charge / EFG — Coulomb;
 $(3\cos^2\theta - 1)/r^3$

bond anisotropy — McConnell
 $\Delta\chi (3\cos^2\theta - 1)/r^3$

H-bond — geometric · Larsen (ppm)

π system — quadrupole

all are projections of the one D_{ab} .

terms θ — angle from the nucleus to the source (ring axis, bond axis, or charge);

r — that nucleus-to-source distance;

$\Delta\chi$ — the bond's susceptibility anisotropy (how directional its magnetic response is).